

Superstore Business Intelligence Dashboard Using Power BI

Project Overview

This project analyzes the superstore sales dataset using Power BI to provide insights into sales performance, profitability, customer behavior, and product performance. The dashboard enables business users to monitor KPIs, identify trends, and make data-driven decisions through interactive visualizations and slicers.

Business Objectives

- Monitor overall sales and profitability
- Analyze sales trends over time
- Identify top-performing products and customers
- Evaluate regional performance
- Understand customer segments.
- Support strategic business decisions

Tools Used

- Power BI
- Power Query
- Dax
- Star Schema Data model

Dashboard Pages

1. Executive Overview

The Executive Overview provides a high-level summary of the business's overall performance. It highlights key performance indicators (KPIs) such as Total Sales, Total Profit, Total Orders, Total Customers, and Total Quantity Sold. The page also presents sales trends, regional performance, product category performance, and customer segment distribution, enabling executives to assess business health quickly.

2. Sales Analysis

The Sales Analysis Dashboard focuses on sales performance across different dimensions, including time, region, category, sub-category, segment, and shipping mode. It helps identify sales trends, top-performing regions, and product categories while providing insights into customer purchasing patterns.

3. Customer Analysis

The Customer Analysis dashboard evaluates customer purchasing behavior and engagement. It highlights top customers by sales and order volume, compares customer segments, and measures customer purchasing frequency, helping businesses identify their most valuable customers.

4. Profitability Analysis

The Profitability Analysis Dashboard examines business profitability by analyzing profit trends across regions, categories, products, and customer segments. It also evaluates the relationship between discounts and profitability to support pricing and discount strategies.

5. Product Analysis

The Product Analysis dashboard provides insights into product performance by examining sales, profit, and quantity sold across products and categories. It

identifies top-performing products and supports inventory planning and product management decisions.

Key Business Insights & Recommendations

Executive Overview

Key Insights

- The business generated approximately **\$2.30M** in total sales from over **5,000 orders** and **793 customers**.
- The **Consumer** segment contributed more than half of total sales, making it the largest customer segment.
- The **West** region recorded the highest sales performance, while the **South** region generated the lowest sales.
- Sales showed an upward trend toward the end of the year, indicating stronger business performance in the final quarter.
- **Technology** emerged as the highest-selling product category.

Recommendations

- Continue investing in the Consumer segment through targeted marketing campaigns.
- Develop strategies to improve sales performance in underperforming regions, particularly the South.
- Prepare for increased demand during peak sales periods by optimizing inventory and staffing.
- Expand the product portfolio within the Technology category to maximize revenue opportunities.

Sales Analysis

Key Insights

- Monthly sales followed an overall positive trend, with noticeable increases toward year-end.
- The West and East regions consistently outperformed other regions in sales.
- Technology generated the highest sales among all product categories.
- Consumer customers accounted for the largest share of overall sales.
- Standard Class shipping was the most frequently used shipping mode.

Recommendations

- Strengthen sales campaigns during peak months to maximize revenue.
- Improve marketing efforts in lower-performing regions.
- Continue promoting high-demand technology products.
- Evaluate shipping operations to maintain efficient delivery performance.

Customer Analysis

Key Insights

- A small group of customers generated a significant portion of total sales.
- Consumer customers placed the highest average number of orders.
- Customer purchasing activity varied considerably across different customer segments.
- High-value customers demonstrated strong purchasing consistency.

Recommendations

- Introduce customer loyalty and retention programs for top-performing customers.
- Develop personalized promotions targeting high-value customers.

- Encourage repeat purchases through exclusive offers and reward programs.
- Improve engagement strategies for lower-performing customer segments.

Profitability Analysis

Key Insights

- The business generated approximately **\$286K** in total profit with a profit margin of about **12.5%**.
- Technology delivered the highest overall profit among product categories.
- The West region contributed the highest profit.
- Higher discount levels generally corresponded with lower profitability.
- Certain products generated significantly higher profits than others.

Recommendations

- Review discount policies to protect profit margins.
- Prioritize sales of high-profit products.
- Increase focus on profitable product categories such as Technology.
- Improve profitability in lower-performing regions through targeted pricing strategies.

Product Analysis

Key Insights

- Product sales were concentrated among a relatively small number of top-performing products.
- Office Supplies recorded the highest quantity sold, reflecting strong customer demand.
- Several products generated high sales volumes but comparatively lower profits.

- Average sales varied considerably across individual products.

Recommendations

- Maintain sufficient inventory levels for best-selling products.
- Reassess pricing strategies for products with high sales but low profitability.
- Promote high-margin products through targeted marketing campaigns.
- Periodically review product performance to optimize the product portfolio.